

COMPUTER SOFTWARE

Software refers to various kinds of programs used to operate computers and related devices to causes the Hardware to function in the desired way. The software needs to be accessed before it can be used. There are many terms used for the process of accessing software including running, executing, starting up, opening, and others. The basic difference between the Hardware and Software is just the same as that exists between TV and TV studio. Without a TV studio (software) from where the programs are telecast, the TV (Hardware) is a dead machine.

Software Categories: Software can be classified into two categories:

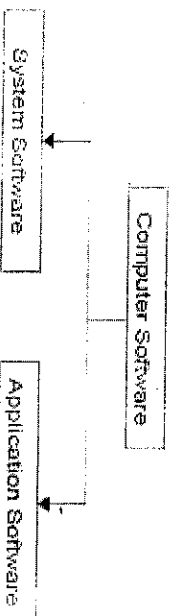


Fig. 29: Software Categories

System Software: System software is the collection of programs designed to **operate, control**, and extend the **processing** capabilities of the computer. These programs provide the environment to run the application programs and are a necessity for proper functioning. Users do not interact with the system software as it works in the background. **The three major classes of system software are;**

1. Operating System
2. Translators
3. Utility programs

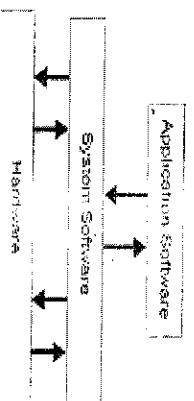


Fig. 30: System Software

Utility Software: Utility software, designed to manage the operating system, support the computer hardware, and directly performing services that benefit ordinary users. Utility software helps to manage, maintain, and control computer resources. Some have been integrated into Operating systems such as file manager, disk cleaner, backup, disk partitioner, launcher applications, network managers, etc.

Antivirus software, as the name suggests, helps to protect a computer system from viruses and other harmful programs.

Backup software helps in the creation of a backup of the files on your hard disk drive of a computer. Backup software helps you copy the most important files on a disk and use it to restore the entire disk in the event of data failure of the selected file in the event of accidental deletion.

Translators: Computers can understand instructions only when they are written in their language, the machine language. Translator a program designed to translate instructions written in any other language into machine language so that computers can execute them. Their translators are; compilers, assemblers, and Interpreters.

7. OPERATING SYSTEM (OS):

The most important type of system software is the operating system. An operating system is an integrated set of specialized programs that are used to manage the overall operations of a computer. It serves as the interface between the computer hardware, the software, and the end-user, and allows a user to run other applications on a computing device. The primary goals of the O. S. are mainly to optimize resources and providing a user-friendly environment. It acts as manager of resources such as CPU time, memory space, file storage, I/O devices and allocates resources among the various programs optimally competing for them. Every computer must have an operating system to run other programs and ensure that available resources are equally shared. Examples of Operating Systems are Mac-OS, MS-DOS, UNIX, Windows O.S. LINUX, Apple Macintosh, and Android O.S.

Windows O.S. Is an operating system with a graphical user interface (icons, pictures, and menus)

Instead of text; uses a mouse as well as a keyboard as an input device)

The Versions are Windows 98, Windows Me, Windows NT, Windows 2000, Windows XP, Windows Server 2003, Windows Vista, Windows 7, etc.

O. S. programs: O. S. consists of several programs. They are categorized into control programs for managing resources and service programs for providing services.

Scheduling strategies

While executing a job, the OS reads the next job from memory into a storage area on the disk and outputs a printout of a previous job from disk to the printer. Spooling allowed the CPU to choose a particular job for execution leading to the concept called **Job Scheduling**. Scheduling is the method by which processes are given access to system resources. The strategies used are:

1. **Multiprogramming system:** In multiprogramming, memory is divided into many partitions.

The O. S. keeps two or more processes that are runnable in memory simultaneously by sharing a single processor. It selects a job from the ready state and starts executing it. When that job needs to wait for some external event the CPU is switched to another job to maximize CPU usage(eliminate idle time).

Multiprogramming allows many programmers to load their programs in different partitions.

2. **Time-sharing:** Multiprogramming was followed by the time-sharing concept. In Time-sharing, the running task is required to relinquish the CPU, either voluntarily or by an external event to another task. This is to allow several programs to execute simultaneously. This is known as a multitasking system. It is a method by which multiple processes share common processing resources by switching between them with a fixed time for each program. In the next cycle, the program that had been considered earlier was taken once again. This process continued until all the programs were executed. A computer with a single CPU uses multitasking to solve problems by scheduling which task may run at any given time and when another waiting task gets a turn. The act of reassigning a CPU from one process to another is called a context switch. Context switch involves two processes one leaves the running state and another enters the running state. A computer with more than one CPU is called a multiprocessing system it is where multiple processors run multiple processes at the same time, which creates physical parallelism.

3. **Major Features of the Operating System:** Based on the number of users, and O. S. can be **Single user:** only one user can operate the system at a time e.g. MS-DOS. **Multi-user O.S.:** More than one user can operate the same system simultaneously and is based on the concept of time-sharing e.g. Unix O.S.

Functions of Operating System:

Today, most operating systems perform the following important characteristics:

1. **Process management:** It controls the jobs submitted to the CPU. Several algorithms are used to allocate the job to the processor e.g. First In First Out; based on queuing, Shortest Job First; select the process that has the shortest expected processing time. Round Robin: Jobs are assigned processor time in a circular method; Based on Priority: In this method, each job is assigned a Priority
2. **Memory management:** It Keeps track of the primary memory, to check what part is in use, by whom, what part is not in use, etc. It manages the allocation of main memory and other storage areas to the system programs as well as user programs and data when a process or program requests it.
3. **Device (I/O) management:** It manages the coordination and assignment of different Input and Output devices while one or more programs are executed. Only the O. S. is empowered to instruct the hardware to make an allocation or de-allocation, which process gets the device, when, and for how much time.
4. **File management:** It allows all files to be easily changed and modified through the use of text editors or some other file manipulation routines.
5. **Job scheduling:** It determines and maintains the order in which jobs are to be executed in the computer system.
6. **Interpretation of commands and instructions.**
7. **Facilitates easy communication between the computer system and the computer operator**
8. **Monitor performance:** It supervises records delays between the request for a service and from the system.
9. **Security Management:** The O. S. provides some levels of securities to the user like the file access, System-level, and Network level to safeguard against unauthorized access. To access the files created by other people, you should have the permission, password in a multi-user environment.

APPLICATION SOFTWARE:

An Application Software consists of programs designed to solve a user problem in a particular area. Users interact with the application software while doing their different activities. It is used to accomplish specific tasks rather than just managing a computer system. Such as a program to list stock items for a particular shop, or a program designed to give a forecast of the weather; the list is endless. These programs are written in different languages such as BASIC or C or by using database packages like dBase, Oracle. Example: Payroll, Billing, Railway Reservation, etc. Application software is in turn, controlled by system software that manages hardware devices. Application software is classified into **customized** applications and general-purpose application programs.

The General-Purpose Application Programs: These packages are developed to suit the needs of research workers/scientists in different fields. These packages are categorized as Word processor (typing letters), Spreadsheet (financial info), Database (inventory control and address book), Data Analysis, Graphics program, *Internet Browser*, Email, and many others, etc. The commonly used Application Software packages are a word processor, spreadsheet, database management system, and graphics.

Word processing software. This software enables you to create, edit, format, store and print text and graphics. Word processors are used in offices (to prepare letters, memos, project reports, minutes), in publishing (books), by journalism (media houses to produce newspapers and magazines), and in education (to prepare test and study materials). Some of the commonly used word processors are Microsoft Word, WordStar, WordPerfect, word pad, etc.

Some Basic Features of word processors

Word processors provide the following basic features:

Insert text: Allows you to insert text anywhere in the document.

Delete text: Allows you to erase characters, words, lines, or pages unlike crossing out on paper.

Cut and paste: Allows you to remove a part of the text from one place in a document and insert it somewhere else.

Copy: Allows you to duplicate a section of text.

Page size and margins: Allows you to define various page sizes and margins, and the processor will automatically readjust the text so that it fits.

Search and replace: Allows you to search for a particular word or phrase and replace it with another.

Word wrap: The word processor automatically moves to the next line once the current line is full

Print: Allows you to send a document to a printer to get a hardcopy.

Spreadsheet software is the computer equivalent of a paper worksheet to organize numeric data in a tabular form. It provides many Mathematical, Financial & Statistical functions, used in many businesses, scientific and engineering environments for analyzing data, calculations, organizes data in tabular form, and creation of graphics and charts. **A spreadsheet** consists of a grid made from columns and rows. Examples of spreadsheet programs are: Lotus 1-2-3, Corel's QuattroPro, MS Excel

Database Applications: A Database is a collection of related information about a particular subject (person, place, or thing) stored in a way that can easily be accessed, managed, and updated. Such as Computerized banking system, Automated Teller Machine, Airlines, and Railway reservation system, etc., are some of the database applications. The common **Characteristics of Database include:** Self-describing, Data exists permanently, multiple users can access the same data, and Data security protects the data from unauthorized access with the help of the passwords. Many software is used as Database Management Systems. Example: MS-Access, ORACLE, MySQL, etc.

Relationship between Hardware and software

1. Hardware and software are mutually dependent on each other.
2. Both of them must work together to make a computer produce a useful output.
3. The software cannot be utilized without supporting hardware.
4. Hardware without a set of programs to operate upon cannot be utilized and is useless.